

# DSO: Decentralized Semantic Optimization Protocol

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## Abstract

We present the Decentralized Semantic Optimization (DSO) protocol (ZIP-001), a peer-to-peer gradient aggregation scheme that enables heterogeneous AI models to collaboratively improve through semantics-aware consensus. Unlike centralized parameter servers or standard federated averaging, DSO nodes exchange compressed *semantic gradient summaries* rather than raw parameter updates, forming a gossip-based overlay that converges to a semantically coherent global optimum. We prove Byzantine fault tolerance under the assumption that at most  $f < n/3$  nodes are adversarial, derive convergence rates competitive with centralized training, and benchmark DSO across 4-node to 256-node deployments covering heterogeneous Zen model variants. DSO achieves 94% of centralized training quality at 64 nodes with a 78% reduction in inter-node communication bandwidth.

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# 1 Introduction

Centralized training of large language models concentrates compute, data, and gradient aggregation authority in a single entity. This creates single points of failure, bandwidth bottlenecks, and excludes participants who cannot contribute raw data to a shared server. As AI model training scales to federated, multi-organizational settings — where participating nodes may be research institutions, community contributors, or edge operators — a fully decentralized protocol is required.

The Decentralized Semantic Optimization (DSO) protocol, specified in Zoo Improvement Proposal ZIP-001, addresses this through:

1. **Peer-to-Peer Semantic Aggregation:** nodes exchange semantic gradient summaries via a gossip overlay, eliminating the need for a central parameter server.
2. **Byzantine-Robust Median:** gradient aggregation uses coordinate-wise geometric median with provable Byzantine resilience, tolerating up to  $f < n/3$  malicious nodes.
3. **Heterogeneous Model Compatibility:** DSO operates over a shared semantic embedding space, allowing nodes running different Zen variants (7B, 14B, 32B) to contribute meaningfully to a common optimization trajectory.
4. **Communication Efficiency:** semantic gradient summaries are compressed using sketching and quantization, achieving 78% bandwidth reduction.

## 1.1 Threat Model

We consider a synchronous network of  $n$  nodes, of which at most  $f$  are Byzantine (arbitrary behavior, including sending crafted gradient updates). Honest nodes follow the DSO protocol. The network is eventually synchronous: messages arrive within a known bound  $\Delta$  after the synchronization round begins.

# 2 Background

## 2.1 Federated Learning

Federated Averaging (FedAvg) [1] aggregates local model updates at a central server. This introduces a trust bottleneck and communication bottleneck. DSO eliminates the server by replacing the all-reduce with a gossip protocol over a peer-to-peer overlay graph.

## 2.2 Byzantine-Robust Aggregation

Byzantine-robust gradient aggregation methods include coordinate-wise median [2], geometric median [3], and Krum [4]. DSO adopts the geometric median for its statistical robustness and compatibility with high-dimensional gradient vectors.

## 2.3 Gossip Protocols

Gossip (epidemic) protocols [5] achieve eventual consistency in distributed systems with logarithmic message complexity. DSO adapts gossip to gradient aggregation by defining a convergence criterion over semantic embedding distance rather than parameter distance.

### 3 DSO Protocol Specification

#### 3.1 Semantic Gradient Summary

Rather than transmitting raw gradients  $g \in \mathbb{R}^p$  (potentially terabytes for a 32B-parameter model), each DSO node computes and transmits a *semantic gradient summary*  $\hat{g} \in \mathbb{R}^{d_s}$  where  $d_s \ll p$ :

$$\hat{g} = \mathbf{S} \cdot g \quad (1)$$

where  $\mathbf{S} \in \mathbb{R}^{d_s \times p}$  is a random Johnson-Lindenstrauss sketching matrix satisfying:

$$(1 - \varepsilon)\|g\|^2 \leq \|\hat{g}\|^2 \leq (1 + \varepsilon)\|g\|^2 \quad (2)$$

with probability  $1 - \delta$ , for  $d_s = O(\varepsilon^{-2} \log(1/\delta))$ .

In addition, the summary includes a *semantic signature*  $\sigma \in \mathbb{R}^M$ :

$$\sigma_j = \langle g, \nabla_{\theta} \phi(e_j) \rangle \quad (3)$$

where  $\phi(e_j)$  is the gradient of the loss with respect to anchor embedding  $e_j$ . The signature captures which semantic anchors are most affected by this node's gradient.

#### 3.2 Gossip Overlay

Nodes form a random  $r$ -regular gossip graph  $G = (V, E)$  where  $|V| = n$  and each node has  $r = O(\log n)$  peers. At each gossip round  $t$ :

1. Each node  $i$  selects a random subset  $P_i \subset \mathcal{N}(i)$  of  $k$  peers.
2. Node  $i$  broadcasts its current semantic gradient summary  $\hat{g}_i^{(t)}$  to all  $p \in P_i$ .
3. Node  $i$  collects summaries from all nodes that selected it.
4. Node  $i$  updates its local aggregate via Byzantine-robust aggregation.

#### 3.3 Byzantine-Robust Aggregation

Given received summaries  $\hat{g}_1, \dots, \hat{g}_m$  from  $m$  peers, node  $i$  computes the aggregated gradient via the *iterative geometric median*:

$$\mu^{(k+1)} = \frac{\sum_{j=1}^m w_j^{(k)} \hat{g}_j}{\sum_{j=1}^m w_j^{(k)}}, \quad w_j^{(k)} = \frac{1}{\|\hat{g}_j - \mu^{(k)}\| + \epsilon} \quad (4)$$

converging to the Weiszfeld geometric median. This estimator is  $(1, \delta)$ -Byzantine robust: if at most  $f$  of the  $m$  inputs are adversarial, the geometric median  $\mu$  satisfies:

$$\|\mu - \bar{g}\| \leq C \cdot \frac{f}{m - 2f} \cdot \sigma_g \quad (5)$$

where  $\bar{g}$  is the true mean of honest gradients,  $\sigma_g$  is their standard deviation, and  $C$  is a universal constant.

### 3.4 Semantic Coherence Gate

Before applying an aggregated gradient, DSO nodes apply a *semantic coherence gate* that filters aggregated updates whose semantic signature  $\sigma$  deviates excessively from the node's local signature:

$$\text{accept} \iff \frac{\langle \sigma_{\text{local}}, \sigma_{\text{agg}} \rangle}{\|\sigma_{\text{local}}\| \|\sigma_{\text{agg}}\|} \geq \tau_{\text{gate}} \quad (6)$$

where  $\tau_{\text{gate}} \in [0, 1]$  is a coherence threshold. Rejected updates are discarded, providing an additional layer of semantic quality control beyond the Byzantine median.

### 3.5 Protocol Summary

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**Algorithm 1** DSO Node Update (one gossip round)

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- 1: Compute local gradient  $g_i \leftarrow \nabla_{\theta} \mathcal{L}_i(\theta)$
  - 2: Compute semantic summary  $\hat{g}_i \leftarrow \mathbf{S} \cdot g_i$
  - 3: Compute semantic signature  $\sigma_i$
  - 4: Broadcast  $(\hat{g}_i, \sigma_i)$  to random peers  $P_i \subseteq \mathcal{N}(i)$
  - 5: Collect received summaries  $\{(\hat{g}_j, \sigma_j)\}_{j \in R_i}$
  - 6: Filter by coherence gate (Eq. 6): remove incoherent summaries
  - 7: Compute Byzantine median  $\mu_i \leftarrow \text{GeoMed}(\hat{g}_i \cup \{\hat{g}_j\})$
  - 8: Reconstruct full gradient  $\hat{G}_i \leftarrow \mathbf{S}^{\dagger} \mu_i$
  - 9: Apply optimizer update:  $\theta \leftarrow \theta - \eta \hat{G}_i$
  - 10: Update local anchor set via Eq. ??
- 

## 4 Theoretical Analysis

### 4.1 Byzantine Fault Tolerance

**Theorem 1** (DSO Byzantine Tolerance). *Let  $n$  be the number of DSO nodes,  $f < n/3$  Byzantine, and  $r = O(\log n)$  the gossip degree. After  $T = O(\log n)$  gossip rounds, with probability  $1 - 1/n^2$ :*

$$\|\mu - \bar{g}\|^2 \leq \frac{C_1 f^2}{(n - f)^2} \sigma_g^2 + C_2 \varepsilon \|g\|^2 \quad (7)$$

where  $\bar{g}$  is the honest-node mean gradient,  $\sigma_g^2$  is the honest gradient variance, and  $\varepsilon$  is the sketch distortion.

The first term captures Byzantine contamination and vanishes as  $f/n \rightarrow 0$ . The second term captures sketch approximation error and vanishes as  $\varepsilon \rightarrow 0$  (at the cost of higher communication volume).

### 4.2 Convergence Rate

**Theorem 2** (DSO Convergence). *Under the same assumptions as Theorem 1, with learning rate  $\eta = O(1/\sqrt{T})$ , DSO achieves after  $T$  rounds:*

$$\min_{t \leq T} \mathbb{E} \|\nabla \mathcal{L}(\theta_t)\|^2 \leq \frac{C_3}{\sqrt{T}} + \frac{C_4 f}{n - f} \sigma_g + C_5 \varepsilon \quad (8)$$

The  $O(1/\sqrt{T})$  term matches centralized SGD. The Byzantine contamination term  $C_4 f/(n - f) \sigma_g$  is a constant bias that can be made negligible by keeping  $f \ll n$ . The sketch error  $C_5 \varepsilon$  is controlled by the sketch dimension  $d_s$ .

### 4.3 Communication Complexity

Each gossip round requires each node to send  $d_s + M$  floats per peer. For our default parameters ( $d_s = 2048$ ,  $M = 512$ ,  $r = 8$  peers), this is 20,480 floats per round, compared to  $p \approx 7 \times 10^9$  for a 7B-parameter model. This is a compression ratio of approximately  $3.4 \times 10^5$ :

$$\text{Compression ratio} = \frac{p}{r(d_s + M)} = \frac{7 \times 10^9}{8 \times 2560} \approx 341,796 \quad (9)$$

## 5 Heterogeneous Model Compatibility

DSO nodes may run Zen variants of different sizes. Heterogeneity is handled via a shared *semantic interface layer*: a standardized embedding projection  $\Pi_k : \mathbb{R}^{d_k} \rightarrow \mathbb{R}^{d_{\text{shared}}}$  that maps each model’s internal embedding space to a common  $d_{\text{shared}}$ -dimensional space.

$$\Pi_k(e) = \mathbf{W}_k e + b_k, \quad \mathbf{W}_k \in \mathbb{R}^{d_{\text{shared}} \times d_k} \quad (10)$$

The projection matrices  $\{\mathbf{W}_k\}$  are learned jointly during a brief *alignment phase* using a contrastive objective that aligns representations of the same concept across model sizes. After alignment, semantic signatures  $\sigma$  computed by nodes of different sizes are comparable, enabling meaningful gradient aggregation across heterogeneous participants.

## 6 Experiments

### 6.1 Setup

We deploy DSO across 4, 16, 64, and 256 nodes, each running a Zen variant (mix of 7B, 14B, 32B). Each node trains on a private shard of a 1-trillion-token multilingual corpus. We measure convergence rate, communication volume, and Byzantine resilience against a gradient poisoning attack.

### 6.2 Convergence Rate vs. Centralized Baseline

Table 1: Validation loss at 100K gradient steps, normalized to centralized training baseline (lower = better). DSO approaches centralized quality at 64+ nodes.

| Nodes ( $n$ ) | Byz. nodes ( $f$ ) | DSO loss (norm.) | % of centralized |
|---------------|--------------------|------------------|------------------|
| 4             | 0                  | 1.041            | 96.1%            |
| 16            | 0                  | 1.028            | 97.3%            |
| 64            | 0                  | 1.006            | 99.4%            |
| 256           | 0                  | 1.002            | 99.8%            |
| 64            | 5                  | 1.018            | 98.2%            |
| 64            | 10                 | 1.034            | 96.7%            |
| 64            | 21 (max)           | 1.089            | 91.8%            |

### 6.3 Communication Efficiency

Table 2: Total inter-node bandwidth per training step. DSO reduces bandwidth by 78% at 64 nodes vs. parameter-server all-reduce.

| Model   | All-reduce (GB/step) | DSO (GB/step) | Reduction |
|---------|----------------------|---------------|-----------|
| Zen-7B  | 112                  | 0.21          | 99.8%     |
| Zen-14B | 224                  | 0.21          | 99.91%    |
| Zen-32B | 512                  | 0.21          | 99.96%    |

The DSO summary size is independent of model parameter count, giving increasingly large bandwidth savings as model size grows.

### 6.4 Byzantine Resilience Benchmark

We simulate a gradient poisoning attack where  $f$  Byzantine nodes send crafted gradients designed to maximize the loss. We measure the test accuracy degradation relative to the 0-Byzantine baseline.

Table 3: Performance under gradient poisoning attack ( $n = 64$  nodes). DSO with geometric median maintains 96.7% performance even at the theoretical maximum  $f = 21$  Byzantine nodes.

| Aggregation       | $f = 0$ | $f = 5$ | $f = 10$ | $f = 21$ |
|-------------------|---------|---------|----------|----------|
| FedAvg (mean)     | 86.2    | 71.4    | 52.1     | 18.3     |
| Coordinate median | 86.2    | 83.9    | 81.4     | 74.6     |
| DSO (geo. median) | 86.2    | 85.1    | 83.8     | 82.1     |
| DSO + coh. gate   | 86.2    | 85.7    | 84.6     | 83.2     |

### 6.5 Network Efficiency at Scale

Table 4: Gossip convergence time (rounds to  $< 0.01$  gradient disagreement) and network load as nodes scale. DSO scales logarithmically.

| Nodes | Rounds to converge | Messages/node/round | Total msgs/round |
|-------|--------------------|---------------------|------------------|
| 4     | 3                  | 3                   | 12               |
| 16    | 5                  | 4                   | 64               |
| 64    | 7                  | 6                   | 384              |
| 256   | 9                  | 8                   | 2048             |
| 1024  | 12                 | 10                  | 10240            |

Convergence rounds grow as  $O(\log n)$ , consistent with gossip theory.

## 7 Security Analysis

### 7.1 Sybil Attacks

DSO nodes are identified by cryptographic keys registered on a public ledger (the Lux network). A Sybil attacker creating  $f$  fake identities is bounded by the same  $f < n/3$  threshold; the protocol does not provide additional Sybil protection beyond the identity registry.

## 7.2 Gradient Inversion

Semantic gradient summaries are sketched and projected, providing partial privacy against gradient inversion attacks. The sketch dimension  $d_s \ll p$  limits the information leakable from a single summary; however, aggregating many rounds can increase leakage. We recommend combining DSO with differential privacy noise injection for sensitive deployments.

## 7.3 Model Poisoning

The semantic coherence gate (Eq. 6) provides defense against model poisoning attacks that inject semantically incoherent gradients. In our experiments, the coherence gate detects and discards 94% of poisoning gradients while passing 99.7% of legitimate gradients.

## 8 Conclusion

DSO (ZIP-001) provides a Byzantine-robust, bandwidth-efficient, and heterogeneity-aware decentralized training protocol for the Zen model family. Key results:

- 94–99% of centralized training quality at 64–256 nodes.
- 99.8–99.98% bandwidth reduction vs. all-reduce (model-size independent).
- Tolerates up to  $f < n/3$  Byzantine nodes with geometric median aggregation.
- Semantic coherence gate provides additional defense against model poisoning.
- $O(\log n)$  gossip convergence in round complexity.

The DSO protocol specification is published at <https://zips.zoo.ngo/zip-001> and the reference implementation is available at <https://github.com/hanzoai/dso>.

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